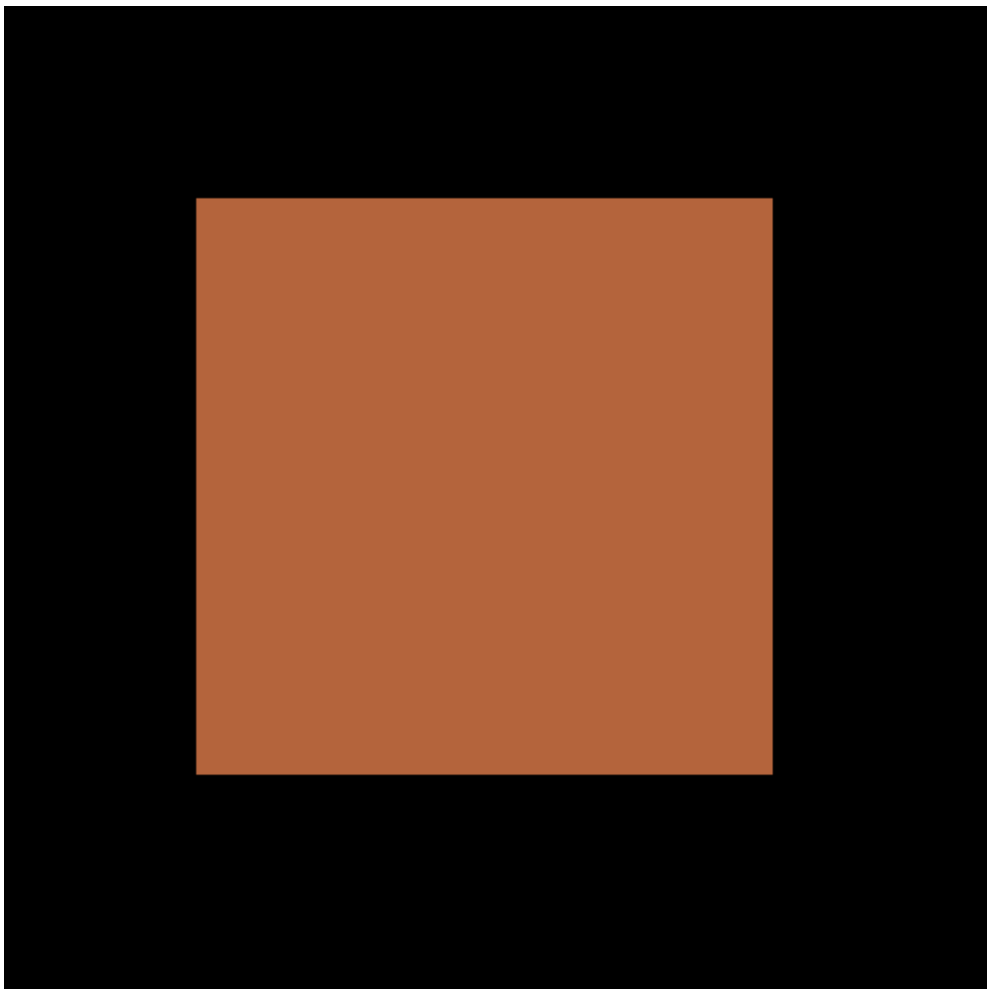


Aakhi — Retinal Image Analysis Report

Phase 1 Report (DR, OD/OC, Lesions, Glaucoma)

Patient Name	Test Patient	Doctor	Dr. Test Doctor
Age	45	Date	08 May 2026, 10:12 AM
Gender	Male	Eye Examined	Right Eye (OD)
Report ID	0F7FD828	Institution	IIT Bhubaneswar — Eye AI Lab

Fundus Image



Original fundus photograph

DR Grading

Classification: **Mild DR** (Level 1/4)

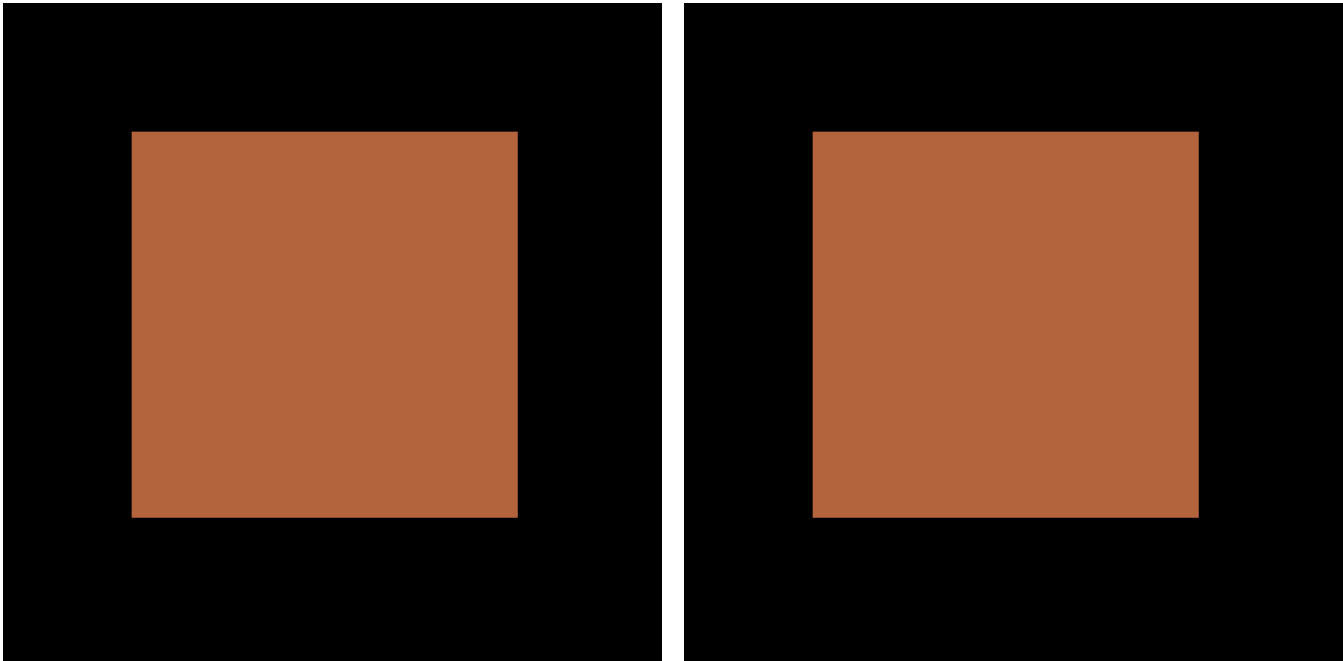
Clinical Description: Mild non-proliferative DR. Presence of at least one microaneurysm. Regular monitoring recommended.

DR Severity Level	1 / 4 — Mild DR
Grading Scale	ETDRS / International Clinical DR Disease Severity Scale
Model	EfficientNet-B6 (fine-tuned, 5-class)

Recommended Action	Follow-up in 6–12 months. Optimise glycaemic and blood pressure control.
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Optic Disc / Cup Segmentation

Color Key: Green=Optic Disc | Blue=Optic Cup | Quadrant Analysis (S=Superior, I=Inferior, N=Nasal, T=Temporal)



Segmentation (Green=Disc, Blue=Cup) Overlay with quadrant lines (S/I/N/T)

Morphometric Measurements

Vertical CDR (vCDR)	0.450 — CDR within normal limits (< 0.5). Low suspicion for glaucomatous optic neuropathy.
Horizontal CDR (hCDR)	0.400
Area-based CDR	0.380
Optic Disc — Vertical Ø	400 px
Optic Disc — Horizontal Ø	420 px
Optic Cup — Vertical Ø	180 px
Optic Cup — Horizontal Ø	168 px
Disc Area (px²)	125664
Cup Area (px²)	25447

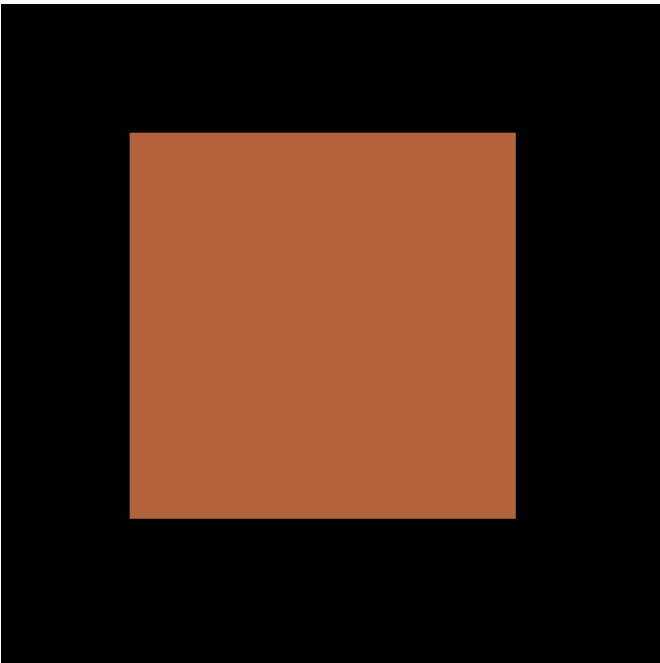
Neuroretinal Rim Analysis — ISNT Rule

Normal ISNT rule: Inferior rim Superior Nasal Temporal

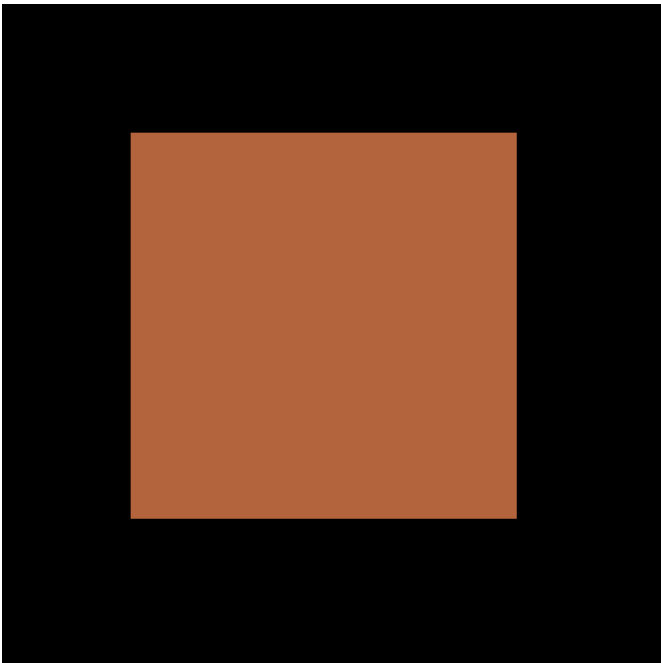
Superior Rim	28.2% of total rim area
Inferior Rim	29.1% of total rim area
Nasal Rim	21.4% of total rim area
Temporal Rim	21.3% of total rim area
ISNT Rule	Normal (ISNT rule satisfied)

Lesion Detection

Color Key: Red=Hard Exudates | Green=Hemorrhages | Blue=Microaneurysms | Yellow=Soft Exudates



Lesion segmentation (colour-coded by type)



Overlay on original fundus

Detected Lesion Areas (pixels²)

Hard Exudates (HE)	500 px ² — Lipid deposits from leaking vessels
Haemorrhages (HEM)	200 px ² — Intraretinal bleeding
Microaneurysms (MA)	100 px ² — Small capillary bulges
Soft Exudates (SE)	50 px ² — Cotton-wool spots
Model	UNet + FIAM attention (IDRiD dataset)

Glaucoma Grading

Glaucoma Severity Grade: No Glaucoma

Clinical Interpretation: No evidence of glaucomatous optic neuropathy.

Model	SE-ResNet101 (3-class glaucoma grading)
Classification	No Glaucoma
Corroborating CDR	vCDR = 0.450 (suspicion threshold: 0.65)
ISNT Rule Violation	No

Clinical Summary

DR Grading	Mild DR (Level 1/4)
Glaucoma Grading	No Glaucoma
Vertical CDR	0.450 — Risk: Low
ISNT Rule	Normal
Lesion Load	HE: 500 HEM: 200 SE: 50 px ²

Recommendations

- DR follow-up: Follow-up in 6–12 months. Optimise glycaemic and blood pressure control.

Report ID: 0F7FD828 | 08 May 2026, 10:12 AM

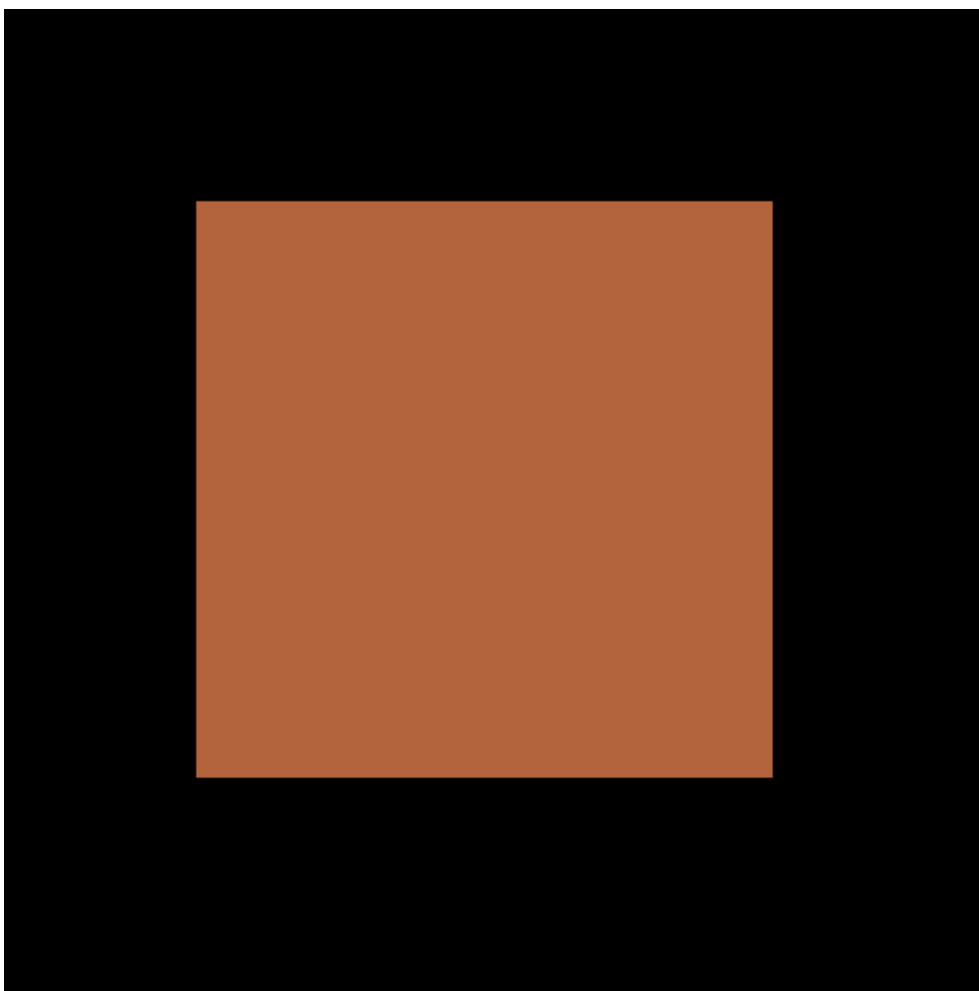
Generated by Aakhi Retinal AI

This report is generated by the Aakhi AI system for clinical reference only. All findings must be verified by a qualified ophthalmologist before treatment decisions.

ଆଁଖି — ରଟେନାଲ ଚିତ୍ରର ବଶିଳଷେଷ ରିପୋର୍ଟ

ପ୍ରଥମ ପର୍ୟାୟ ରିପୋର୍ଟ (, /; କ୍ଷତ, ଗ୍ଲୁକୋମା)

ରୋଗୀର ନାମ	Test Patient	ଡାକ୍ତର	Dr. Test Doctor
ବୟସ	45	ତାରିଖ	08 May 2026, 10:12 AM
ଲିଙ୍ଗ	Male	ପରୀକ୍ଷିତ ଚକ୍ଷୁ	Right Eye (OD)
ରିପୋର୍ଟ	0F7FD828		IIT Bhubaneswar — Eye AI Lab



Original fundus photograph

ଡାୟାବଟିକ ରଟେନୋପ୍ୟାଥ ଗ୍ରେଡିଂ

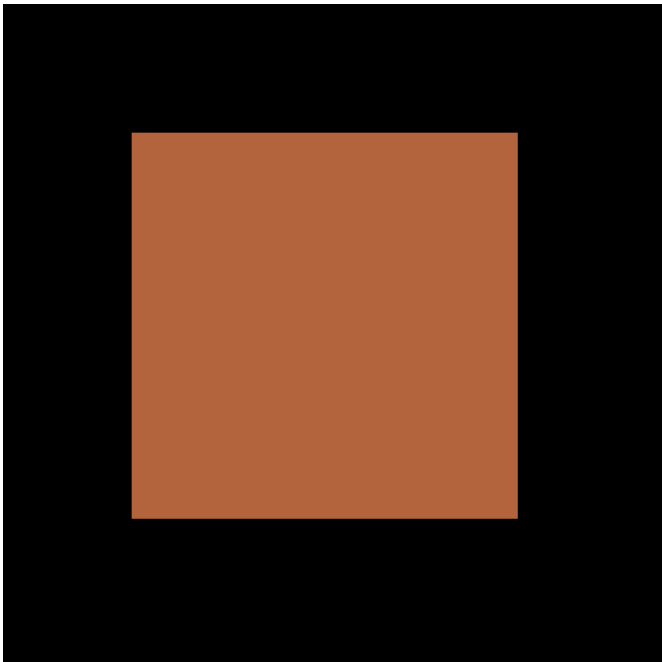
Classification: Mild DR (Level 1/4)

Clinical Description: Mild non-proliferative DR. Presence of at least one microaneurysm. Regular monitoring recommended.

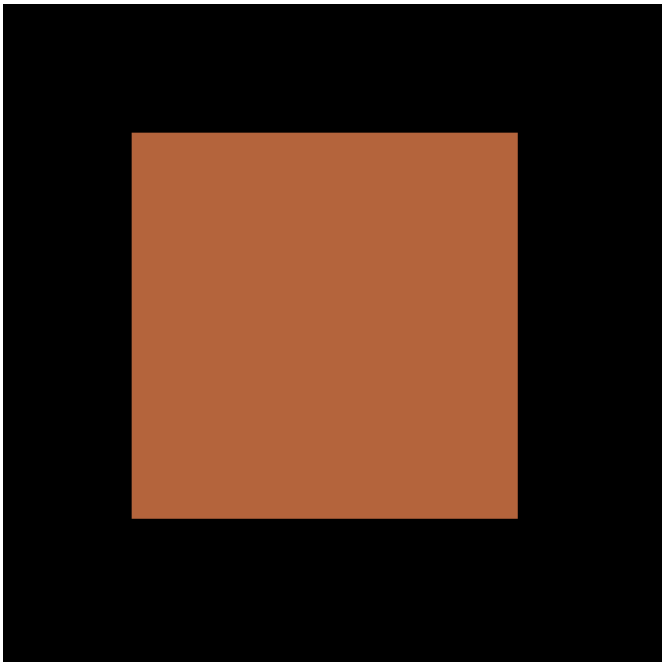
DR Severity Level	1 / 4 — Mild DR
Grading Scale	ETDRS / International Clinical DR Disease Severity Scale
Model	EfficientNet-B6 (fine-tuned, 5-class)
Recommended Action	Follow-up in 6–12 months. Optimise glycaemic and blood pressure control.

অপটিকি ডিস্ক / কাপ সগেমেন্টেশন

রঙের চাবিকাঠি: সবুজ=অপটিকি ডিস্ক | নীল=অপটিকি কাপ | চতুর্ভাগ বিশ্লেষণ (=উপরি, =নম্নি, =নাক, =টেম্পোরাল)



Segmentation (Green=Disc, Blue=Cup)



Overlay with quadrant lines (S/I/N/T)

Morphometric Measurements

Vertical CDR (vCDR)	0.450 — CDR within normal limits (< 0.5). Low suspicion for glaucomatous optic neuropathy.
Horizontal CDR (hCDR)	0.400
Area-based CDR	0.380
Optic Disc — Vertical Ø	400 px
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Disc Area (px ²)	125664
Cup Area (px ²)	25447

Neuroretinal Rim Analysis — ISNT Rule

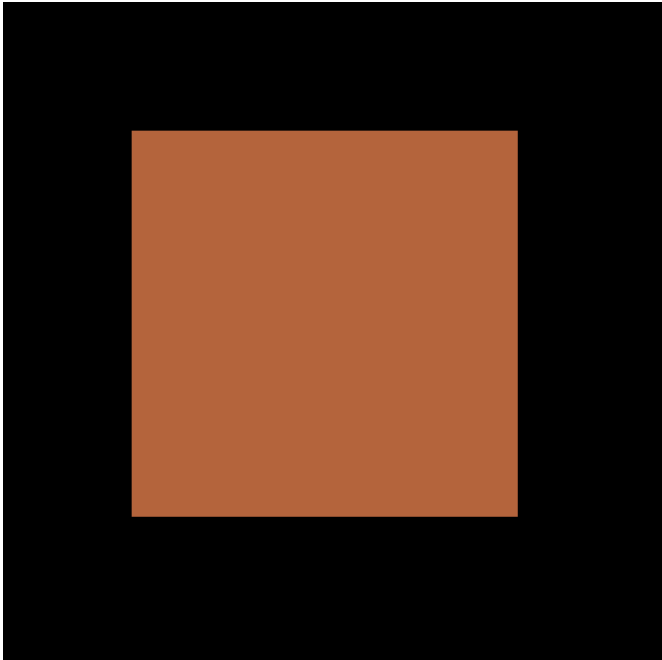
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Superior Rim	28.2% of total rim area
Inferior Rim	29.1% of total rim area
Nasal Rim	21.4% of total rim area
Temporal Rim	21.3% of total rim area

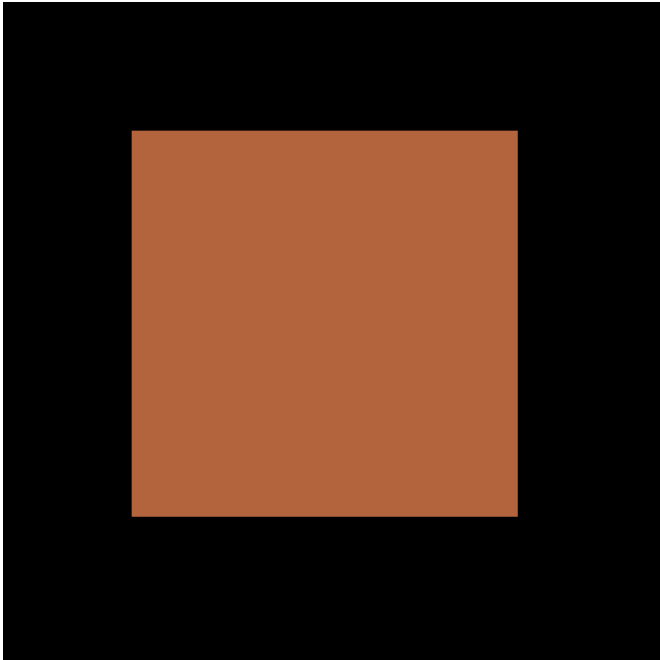
ISNT Rule	(ISNT)
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ক্ষত সনাক্তকরণ

রঙের চাবিকাঠি: লাল=কঠনি এক্সুডেট | সবুজ=রক্তক্ষরণ | নীল=মাইক্রোঅ্যানডিরজিম | হলুদ=নরম এক্সুডেট



Lesion segmentation (colour-coded by type)



Overlay on original fundus

Detected Lesion Areas (pixels²)

Hard Exudates (HE)	500 px ² — Lipid deposits from leaking vessels
Haemorrhages (HEM)	200 px ² — Intraretinal bleeding
Microaneurysms (MA)	100 px ² — Small capillary bulges
Soft Exudates (SE)	50 px ² — Cotton-wool spots
Model	UNet + FIAM attention (IDRiD dataset)

গ্লুকোমা গ্রডেং

Glaucoma Severity Grade: No Glaucoma

Clinical Interpretation: No evidence of glaucomatous optic neuropathy.

Model	SE-ResNet101 (3-class glaucoma grading)
Classification	No Glaucoma
Corroborating CDR	vCDR = 0.450 (suspicion threshold: 0.65)
ISNT Rule Violation	No

ক্লিনিক্যাল সারসংক্ষেপ

DR Grading	Mild DR (Level 1/4)
Glaucoma Grading	No Glaucoma
Vertical CDR	0.450 — Risk: Low
ISNT Rule	Normal
Lesion Load	HE: 500 HEM: 200 SE: 50 px ²

সুপারিশ

- DR follow-up: Follow-up in 6–12 months. Optimise glycaemic and blood pressure control.

Report ID: 0F7FD828 | 08 May 2026, 10:12 AM

অটোম্যাটিক দ্বারা তৈরি

এই রিপোর্টটি শুধুমাত্র ক্লিনিক্যাল রফোরেন্সের জন্য অটোম্যাটিক সিস্টেমে দ্বারা তৈরি করা হয়েছে। চিকিৎসা সংক্রান্ত সিদ্ধান্ত নেওয়ার আগে একজন যোগ্য চক্ষু বিশেষজ্ঞ দ্বারা সমস্ত ফলাফল যাচাই করা আবশ্যিক।